

Driving on Registers, Reasoning on Risk: Risk-Aware Occupancy for Register-Based End-to-End Autonomous Driving

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Abstract

Multimodal trajectory prediction improves behavioral coverage in end-to-end autonomous driving, but existing methods remain limited by sparse scene representations. Incomplete evidence leads to low-quality candidate generation and unreliable ranking among geometrically similar trajectories. On a register-based baseline, bad and poor candidates constitute 19.74% of the candidate set, while the oracle-best candidate ranks only 33.9th on average. We propose RRDrive, which introduces risk-aware occupancy as a dense, temporally aligned, and trajectory-queryable representation. Its global structure guides high-quality multimodal generation, while candidate-conditioned risk queries support fine-grained selection. We further construct RiskOcc-NAVSIM with automatic risk annotations. RRDrive achieves a selected-trajectory PDMS of **0.951**, representing a 1.5% relative improvement over the baseline (0.937), and improves the average candidate PDMS by 7.7%. In challenging scenes, it improves candidate PDMS by 30.2% and increases the Spearman correlation among good candidates by 0.41, from 0.26 to 0.67. To move beyond this oracle setting, we further develop an external RiskOcc predictor, a perception module that estimates risk-aware occupancy directly from sensor inputs. The competitive performance validates the representation’s feasibility.

Introduction

End-to-end autonomous driving must produce a single executable trajectory from uncertain observations. Multimodal trajectory prediction improves robustness by preserving multiple plausible driving behaviors before final selection. However, existing methods still suffer from low-quality candidate generation and unreliable trajectory ranking. We argue that these failures are not inherent to multimodality, but largely stem from sparse scene representations. Agent- and map-based features provide compact structural information, yet omit continuous evidence about unstructured obstacles, road boundaries, uncertain regions, and future interactions. Consequently, they provide insufficient global guidance for candidate generation and inadequate local risk evidence for a risk-aware selector.

We analyze this limitation on baseline, a register-based planner with separate trajectory generation and scoring (Kirby et al. 2026). As shown in Figure 1, a single complex driving scene exhibits two coupled failure modes. The generator outputs abundant unsafe, off-route, or ineffective

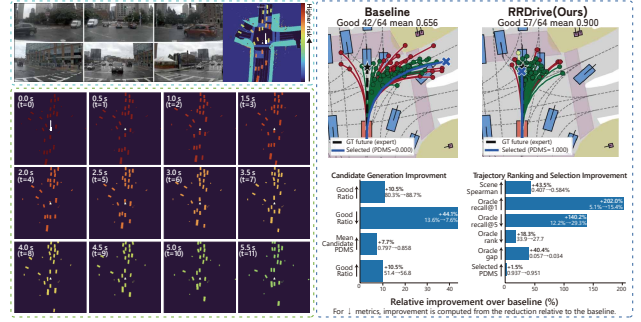


Figure 1: Motivation and overall performance of RRDrive. Register-based baselines face both candidate generation waste and ranking errors in complex scenes, driven by unmodeled local future risks. Left: Multi-view inputs and temporal evolution of risk-aware occupancy. Right: Trajectory comparison and quantitative improvements validate our dense trajectory-queryable risk representation.

candidates that exhaust the limited candidate budget. Meanwhile, geometrically similar good trajectories form dense bundles; their ground-truth oracle PDMS differs from localized future risks, leading to mis-ranking by the predicted scorer. A missing good candidate cannot be recovered by the scorer, while incorrect ranking discards already-available high-quality solutions. These observations suggest that multimodal generator-selector architectures require a dense representation supporting both global risk reasoning and trajectory-specific evaluation.

The risk-aware occupancy, an explicit, dense, and trajectory-queryable representation in the ego-centric BEV form, proposed by (Chen et al. 2024). It unifies map-derived static risks and dynamic occupancy risks within a common field. Its global structure exposes feasible and hazardous regions, enabling the generator to allocate candidates toward high-quality behaviors. Its local values can be queried along each trajectory to form a candidate-specific risk profile, revealing differences in collision exposure, road-boundary risk, and future interactions. Risk-aware occupancy therefore provides the complementary information required for both candidate generation and fine-grained selection.

Building upon this representation, we present a

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Risk-Register E2E framework driven by global-local risk-aware occupancy cues (RRDrive) (Figure 2), a multi-modal end-to-end planner featuring global risk fusion and trajectory-conditioned risk extraction. Global risk fusion incorporates the complete risk field into scene features to improve candidate generation. Trajectory-conditioned extraction samples local risks along each proposal and augments the scorer with explicit risk evidence. We further introduce a risk-aware occupancy prediction branch to estimate the representation from sensor inputs. Because existing datasets do not provide the required supervision, we develop an automatic annotation pipeline and construct RiskOcc-NAVSIM. We also adopt targeted metrics to separately evaluate candidate quality and ranking reliability.

Experiments on NAVSIM show that RRDrive improves both capabilities. It raises the overall average candidate PDMS from 0.797 to 0.858 and the selected-trajectory PDMS from 0.9367 to 0.9506. In generation-challenging scenes, the average candidate PDMS increases from 0.5395 to 0.7026, while the proportion of good candidates rises from 50.13% to 69.76%. In selection-challenging scenes, the Spearman correlation among good candidates improves from 0.2646 to 0.6745, and oracle-best recall increases from 3.62% to 24.86%. These results demonstrate that explicit risk representation improves both multimodal trajectory quality and best-trajectory selection.

The main contributions are summarized as follows:

- We identify compact scene register representation as the shared bottleneck of multimodal trajectory generation and selection, and exploit risk-aware occupancy for global-local collaborative reasoning: global risk guides feasible candidate allocation, while trajectory-conditioned local risk queries offer discriminative cues for ranking geometrically similar trajectories.
- We propose RRDrive, a register-based end-to-end driving architecture that integrates risk-conditioned candidate generation and trajectory-conditioned risk-aware selection. It substantially improves candidate-pool quality and ranking fidelity, yielding consistent gains in challenging interactive scenarios.
- We build RiskOcc4D-NAVSIM by automated annotation of NAVSIM to furnish ground-truth supervision and an evaluation testbed for risk-aware occupancy. We further develop an external RiskOcc predictor producing lossy risk maps from sensor inputs to validate pipeline robustness under non-oracle perceptual conditions.

Related Work

Existing end-to-end autonomous driving planners fall into three paradigms by trajectory candidate construction and selection: deterministic point estimation, discrete hypothesis scoring, and continuous distribution modeling. Each has clear performance trade-offs and inherent flaws. We take the practical discrete candidate scoring paradigm as baseline, targeting its core defect: implicit modeling of scene-trajectory matching. Represented by UniAD (Hu et al. 2023) and TransFuser (Chitta et al. 2022), deterministic single-trajectory regression adopts end-to-end networks with behavior cloning

to output one trajectory per forward pass. Simple, low-latency and multi-task trainable, it fits real-time onboard deployment. Yet it frames planning as point estimation and cannot preserve multiple multimodal feasible trajectories. In interactive scenarios with diverse plausible maneuvers, it outputs averaged, conservative suboptimal trajectories with weak decision diversity and poor long-tail robustness.

Discrete candidate generation with learnable scoring (VADv2 (Jiang et al. 2026), SparseDrive (Sun et al. 2024)) produces discrete trajectory candidates via sparse queries and structured anchors, then ranks trajectories via a dedicated scoring network. It resolves the diversity deficit of single-trajectory regression and avoids generative models’ iterative sampling cost, balancing multi-hypothesis reasoning and fast inference. Its key bottleneck lies in globally compressed implicit scene features for trajectory evaluation: candidate-environment and interaction boundary matching is only implicitly inferred without explicit modeling or precise querying. This yields abundant low-quality redundant samples and unstable fine-grained ranking among similar high-quality trajectory candidates, capping performance in complex interactive scenarios.

Generative sampling with Best-of-N selection (DiffusionDrive (Liao et al. 2025), GoalFlow (Xing et al. 2025), TrajFlow (Li et al. 2026b)) learns scene-conditioned continuous trajectory distributions to break discrete candidates’ expressiveness limits, generating highly diverse, scene-wide trajectory samples better suited to long-tail complex interactions. DiffusionDrive leverages anchor priors and truncated diffusion sampling to cut sampling costs while enabling multimodal trajectories. Still, iterative sampling brings heavy inference overhead incompatible with onboard real-time demands; random sampling lacks stability, and trajectory quality heavily relies on post-hoc scoring. Without explicit spatiotemporal environmental constraints, Best-of-N filtering fails to accurately distinguish near-identical high-quality candidates, hurting deployment stability.

Method

Figure 2 gives an overview of RRDrive. On top of a register-based generator-selector planner, risk-aware occupancy enters through two pathways: a risk-conditioned generation module that fuses the global risk field into scene features to steer candidate generation, and a risk-aware selection module that reads the field along each candidate to sharpen ranking. A label-construction pipeline supplies the F ground-truth risk representation for training, and an external predictor estimates it from sensors for sensor-only inference. The four subsections below detail these components in turn.

Risk-Conditioned Candidate Generation

A finite budget spent on unsafe or off-route proposals sets a quality floor that no downstream scorer can recover, so this module makes scene-level risk explicit and conditions candidate generation on it. The framework represents planning as scoring a finite candidate set $\mathcal{T} = \{\tau_i\}_{i=1}^N$, where each proposal $\tau_i = (x_{i,t}, y_{i,t}, \theta_{i,t})_{t=1}^{T_p}$ is an ego-centric future trajectory. In the original design, candidates come from

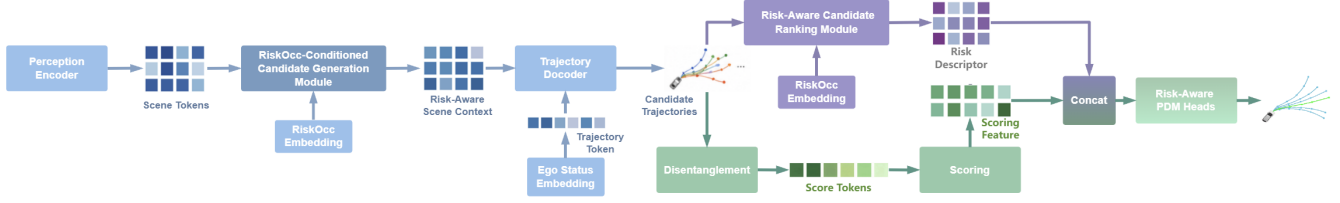


Figure 2: Overall architecture of RRDrive. A risk tokenizer encodes BEV risk-aware occupancy, which is fused with camera scene tokens for risk-conditioned candidate generation and read out along each candidate trajectory for risk-aware selection.

visual scene tokens and trajectory features alone; because this implicit representation cannot separate feasible regions from hazardous ones, the budget leaks toward such proposals, which the explicit RiskOcc field is designed to prevent. Figure 3 illustrates this module.

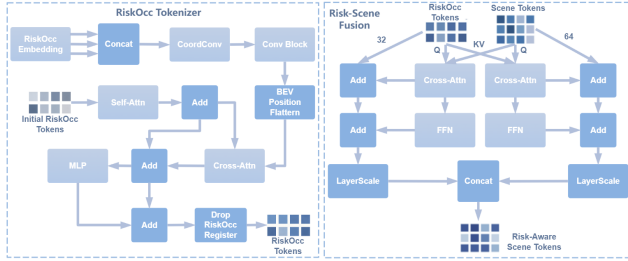


Figure 3: RiskOcc-conditioned candidate generation module. Risk tokens and camera scene tokens are combined by bidirectional cross-attention with LayerScale gating, and the fused context conditions the trajectory decoder so that the finite candidate budget is steered toward feasible, high-quality behaviors.

We use BEV risk-aware occupancy $\mathcal{R} = \{R^s, \{R_t^d\}_{t=1}^{T_r}, R^l\}$, where $R^s \in \mathbb{R}^{H \times W}$ is global scene RiskOcc, $R_t^d \in \mathbb{R}^{C_d \times H \times W}$ is time-indexed dynamic risk with velocity cues, and $R^l \in \mathbb{R}^{H \times W}$ is lane-boundary risk, all in the trajectory frame. Rather than a handcrafted cost map, $\mathcal{R}$ is encoded into a compact token set by a risk tokenizer $\mathcal{E}_\phi$,

$$Z_r = \mathcal{E}_\phi(\text{Concat}(R^s, \{R_t^d\}_{t=1}^{T_r}, R^l)) \in \mathbb{R}^{K \times D}, \quad (1)$$

built from a convolutional BEV stem with coordinate-aware priors, residual channel reweighting, and Perceiver-style cross-attention onto K learnable queries, preserving safety-critical local cues.

Given camera scene tokens Z_c , a risk-aware context is formed by bidirectional cross-attention $\text{CA}(\cdot, \cdot)$,

$$\begin{aligned} \tilde{Z}_c &= Z_c + \gamma_{c \leftarrow r} \text{CA}(Z_c, Z_r), \\ \tilde{Z}_r &= Z_r + \gamma_{r \leftarrow c} \text{CA}(Z_r, Z_c), \end{aligned} \quad (2)$$

whose LayerScale coefficients γ are initialized near zero, so fusion is identity-preserving at initialization and warm-starts stably. The fused context $M = [\tilde{Z}_c, \tilde{Z}_r]$ is cross-attended by the trajectory decoder to produce $\mathcal{T}$, letting global risk structure steer the budget toward feasible behaviors while

suppressing unsafe or off-route proposals. The same M is shared with the selection scorer described next, so risk shapes both candidate representation and final ranking.

Risk-Aware Best-Candidate Selection

Even a strong scorer mis-ranks geometrically similar good candidates when risk stays implicit, so this module hands the scorer a trajectory-conditioned readout of the RiskOcc field to expose the fine progress-risk margin that separates them. For each candidate, scorer attention reads the shared context, $u_i = \text{Attn}(q_i, M)$. Because global tokens alone cannot reveal which risk regions a proposal traverses, we add a trajectory-conditioned local query. At waypoint t , the ego footprint $\mathcal{B}(\tau_{i,t})$ is placed on the BEV grid and bilinearly samples the time-aligned map $\bar{R}_t = [R^s, R_{\alpha(t)}^d, R^l]$, where $\alpha(t)$ aligns waypoint t to dynamic frame t (clamped to $T_r - 1$). For channel c , exposure is a top- k mean over footprint points,

$$\begin{aligned} \mathcal{P}_{i,t,c} &= \text{TopK}_k\{\bar{R}_{t,c}(q) : q \in \mathcal{B}(\tau_{i,t})\}, \\ \rho_{i,t,c} &= \frac{1}{k} \sum_{p \in \mathcal{P}_{i,t,c}} \bar{R}_{t,c}(p), \end{aligned} \quad (3)$$

which retains local peaks that plain averaging would dilute. The per-waypoint sequence is summarized into a fixed descriptor

$$g_i = \text{Concat}(\text{mean}_t \rho_{i,t}, \max_t \rho_{i,t}, \min_t \rho_{i,t}, \rho_{i,T_p}), \quad (4)$$

capturing average exposure, closest-approach risk, low-risk feasibility, and terminal risk. For each submetric $m \in \{\text{NC}, \text{DAC}, \text{DDC}, \text{TTC}, \text{EP}, \text{C}\}$, the descriptor augments only the risk-aware heads $\mathcal{M}_{\text{risk}}$, while the rest use u_i alone,

$$h_i^m = \begin{cases} f_m([u_i, g_i]), & m \in \mathcal{M}_{\text{risk}}, \\ f_m(u_i), & \text{otherwise.} \end{cases} \quad (5)$$

Let $\mathcal{M}_p = \{\text{NC}, \text{DAC}, \text{DDC}\}$, $\mathcal{M}_w = \{\text{TTC}, \text{EP}, \text{C}\}$. Selection keeps the PDM-style weighted-log aggregation,

$$\begin{aligned} s_i &= \sum_{m \in \mathcal{M}_p} w_m \log p_i^m + \log \left(\sum_{m \in \mathcal{M}_w} w_m p_i^m \right), \\ i^* &= \arg \max_i s_i. \end{aligned} \quad (6)$$

This preserves the baseline’s multi-submetric structure and avoids a manual risk penalty. Since simple statistics such as risk mean or max correlate only weakly with true PDMS and safety submetrics, RiskOcc acts not as a linear surrogate but

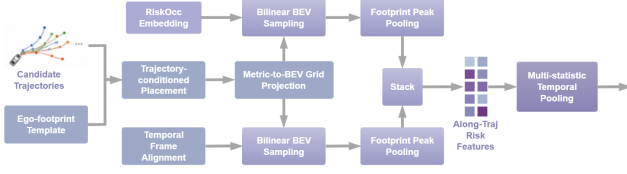


Figure 4: Risk-aware candidate ranking module. For each candidate, the ego footprint samples the time-aligned risk field along the trajectory to form a local risk profile, which augments the risk-sensitive scorer heads and sharpens ranking among geometrically similar good candidates.

as trajectory-aligned nonlinear evidence for the neural scorer. Figure 4 depicts this ranking module.

For the refinement phase, the proposal generator and trajectory decoder are frozen, and only the risk encoder, scorer attention, and the TTC/EP heads are updated. The objective keeps the submetric BCE terms and adds continuous smooth- ℓ_1 calibration (SL_1) for the two risk-sensitive submetrics,

$$\begin{aligned} \mathcal{L} = \sum_m \lambda_m \mathcal{L}_{BCE}(p^m, y^m) + \lambda_{TTC} SL_1(p^{TTC}, y_c^{TTC}) \\ + \lambda_{EP} SL_1(p^{EP}, y_c^{EP}). \end{aligned} \quad (7)$$

Although the non-target heads are frozen, their BCE losses still backpropagate through the shared scorer features, anchoring NC, DAC, DDC, and comfort during refinement. The model thus does not relearn the planner but sharpens its interpretation of the RiskOcc-trajectory relation, concentrating gains in fine-grained ranking among high-quality candidates—particularly TTC-sensitive exposure and progress-aware selection once safety is satisfied—rather than inducing conservative behavior.

Risk-Aware Occupancy Generation

Both pathways above learn from a ground-truth RiskOcc field, so this module constructs that supervision itself as the RiskOcc-NAVSIM labels. We distill dense risk-aware occupancy representation from NAVSIM on a shared ego-centric BEV lattice of $H \times W = 400 \times 400$ grid cells at resolution $\Delta = 0.25$ m (covering $[-50, 50]^2$ m), with risk normalized to $r \in [0, 1]$. An ego-frame point (x, y) maps to the grid cell $(i, j) = (\lfloor (y - y_{\min})/\Delta \rfloor, \lfloor (x - x_{\min})/\Delta \rfloor)$ with $x_{\min} = y_{\min} = -50$ and metric grid cell center p_{ij} . Three pipelines yield complementary targets: a map-derived *lane-marking RiskOcc map*, a point-cloud *global RiskOcc map* refined by that prior, and an annotation-derived *dynamic RiskOcc map*; the last two are stacked into the final label.

Map-Derived Risk-Aware Occupancy From the vectorized HD map, we rasterize a single-channel prior $M \in \mathbb{R}^{H \times W}$, where each grid cell inherits the value of the top-most (highest-priority) layer covering it:

$$M(i, j) = \phi(\ell^*), \quad \ell^* = \arg \max_{\ell \in \mathcal{L}} \pi(\ell) \mathbb{1}[p_{ij} \in \ell], \quad (8)$$

with render priority $\pi(\cdot)$ and coverage indicator $\mathbb{1}[\cdot]$. The layer values ϕ are drivable interior 0, background 0.05,

crosswalk/stop-line/lane boundary 0.1, drivable boundary 0.2, walkway 0.3, and lane-connector centerline -0.1 . M is reused in the global map pipeline.

Global Risk-Aware Occupancy The semantic point cloud is voxelized into a $400 \times 400 \times 16$ grid ($z \in [-2, 2]$); a voxel is occupied if it holds more than two points, taking the majority semantic label V , else background. For each BEV grid cell, we scan its column upward from the ground ($z=0$) for the first salient voxel w^* (neither road nor background) at height index η (larger η = taller). With normalized height $\hat{h} = \Delta\eta/2.5$ and $g = \text{clip}(0.8\hat{h} + 0.2, 0, 1)$, the point-cloud risk is $R_{pc} = 1$ if $V(w^*) = 2$, $R_{pc} = g$ if $V(w^*) \in \{1, 3\}$, and 0 otherwise. We fuse with the lane prior by element-wise maximum ($\vee$) and refine with one grayscale morphological closing (3×3 element B), $R = ((R_{pc} \vee M) \oplus B) \ominus B$. Finally, R is quantized into 11 ordinal classes: class 0 for $R < 0.05$, class 10 for free background ($R=0.05$), and classes 1–9 for monotonically increasing risk over $(0.05, 1]$.

Dynamic Risk-Aware Occupancy We produce a spatio-temporal tensor of $T = 12$ frames (index $k \in \{0, \dots, 11\}$ at $\delta t = 0.5$ s, i.e. $t = k \delta t \in \{0, \dots, 5.5\}$ s), all expressed in the $t=0$ ego frame. Only agents of {vehicles, pedestrians, and bicycles} visible $t=0$ are kept, with short gaps (≤ 2 consecutive missing frames) linearly interpolated. Each agent carries a velocity clamped and normalized into the $t=0$ frame, $\tilde{v} = \text{clip}(v/20, -1, 1)$, $s = \|\tilde{v}\|_2$, and a frame-indexed risk that decays over the horizon and grows with speed:

$$\rho_a^k = \min\left((1 - 0.05k) + 0.05s/\sqrt{2}, 1\right). \quad (9)$$

An agent with center c_a^k , size (ℓ_a, w_a) and heading unit vector $\mathbf{u}_\theta$ occupies the grid cell (i, j) when the grid cell center p_{ij} lies inside its oriented box. Overlaps are resolved by priority, keeping the higher-risk agent and breaking ties by the smaller instance identity, $a^* = \arg \max_a: B_a^k(i, j)=1 (\rho_a^k, -\text{id}_a)$ —so each occupied grid cell stores the winning agent’s risk $\rho_{a^*}^k$ and velocity $(\tilde{v}_x, \tilde{v}_y)$, while free grid cells are zero.

External RiskOcc Predictor

Formulation Injecting a temporal, BEV-based occupancy branch inside our planner would break its lightweight, history-free, PV register-token design, so we keep risk perception external as a standalone predictor that serves only as a probe of the planner’s robustness to lossy predicted RiskOcc. Given a short sensor observation $\mathcal{O}$, this predictor f_θ emits a BEV RiskOcc field of a static map, a lane map, and a dynamic general-movable-object (GMO) sequence with per-cell motion,

$$\begin{aligned} f_\theta(\mathcal{O}) = (\mathbf{R}^s, \mathbf{R}^\ell \in [0, 1]^{H \times W}, \\ \mathbf{R}^d \in [0, 1]^{T \times H \times W}, \mathbf{V} \in \mathbb{R}^{T \times H \times W \times 2}), \end{aligned} \quad (10)$$

with $H = W = 400$, $T = 12$, through a two-stage encoder-decoder $\mathcal{O} \xrightarrow{\Phi} \mathbf{B} \xrightarrow{\Psi} (\cdot)$ with fused BEV $\mathbf{B} \in \mathbb{R}^{256 \times 50 \times 50}$, and $\odot$ denotes the Hadamard product.

Architecture The encoder Φ outputs a BEV feature $\mathbf{B}$ by fusing two streams: an ego-aligned motion history summarized by a selective state-space operator, and a multi-height deformable camera-to-BEV lift following FusionAD (Ye et al. 2023). The streams are merged by residual spatial cross-attention, with modality dropout for robustness to missing sensors. The decoder Ψ upsamples $\mathbf{B}$ with skip fusion and predicts static and lane risk through class-first heads (a class map that conditions a risk regressor).

The dynamic branch operates on a stop-gradient BEV context $\mathbf{P}$. For each future frame t , a per-frame query $\mathbf{q}_t$ drives a deformable forecaster $\mathbf{F}_t = \mathcal{D}(\mathbf{q}_t, \mathbf{P})$; a FiLM module conditioned on an occupancy prior (AdaNorm) refines it into $\tilde{\mathbf{F}}_t$ through a LayerScale residual, and a dense $K = 12$ head maps $\tilde{\mathbf{F}}_t$ to per-level logits $\mathbf{z}_t^d$. The exported dynamic risk is the softmax-class expectation over the discrete levels $\{r_c\}_{c=1}^K$,

$$\mathbf{R}_t^d = \sum_{c=1}^K \text{softmax}(\mathbf{z}_t^d)_c r_c, \quad (11)$$

and the velocity field is gated by the same expectation, $\mathbf{V}_t = \mathbf{R}_t^d \odot \text{Vel}(\tilde{\mathbf{F}}_t)$ with Vel a linear readout. This continuous expectation, rather than a hard threshold, preserves near-frame accuracy, while a clean-label ground truth retaining only $t = 0$ -visible targets suppresses spurious far-frame occupancy.

Experiments

RRDrive closes the two bottlenecks raised in the abstract—the low-quality floor of the candidate set and the unreliable fine-grained ranking of near-identical good candidates—most decisively where the baseline is weakest. Each claim has a dedicated experiment: candidate generation (Table 3), scene-level ranking (Tables 4 and 5), the concentration of both gains in the hardest scenes (Table 6), per-component attribution (Table 2), and benchmark standing (Table 1).

Experimental Setup

We evaluate on NAVSIM *navtest* (12,146 scenes, 64 candidates each), reporting PDMS and its submetrics (NC, DAC, DDC, TTC, EP, C). Candidates are graded by the oracle PDMS q as *bad* ($q < 0.3$), *poor* ($0.3 \leq q < 0.7$), or *good* ($q \geq 0.7$); all methods take the scorer argmax without re-ranking, and the baseline is a faithfully reproduced DriveR under the same protocol. The external RiskOcc predictor is trained offline by

$$\mathcal{L} = \mathcal{L}_s + \frac{1}{2}\mathcal{L}_\ell + \frac{3}{2}\mathcal{L}_d + \frac{1}{5}\mathcal{L}_v + \frac{3}{10}\mathcal{L}_{\text{cam}}, \quad (12)$$

with static/lane heads using focal-CE, Dice, and smooth- ℓ_1 /SSIM, and a dense dynamic $\mathcal{L}_d$ combining CE, Dice, Lovász, a weighted foreground BCE for recall, a false-positive-penalizing Tversky (0.3 FN+0.7 FP), and an AdaNorm auxiliary.

Main Results

Table 1 reports PDMS and its submetrics against recent VLA-based and end-to-end planners on NAVSIM *navtest*. Placing our register-based planner at the top of the end-to-end group confirms that closing the candidate-quality and ranking gaps

Methods	PDMS $\uparrow$	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	Comf. $\uparrow$
Baselines & Oracle						
Human Driver 2024	94.8	100.0	100.0	87.5	100.0	99.9
PDMS-Closed 2023	89.1	94.6	99.8	99.9	86.9	89.9
Ego-stat. MLP 2024	65.6	93.0	77.3	62.8	83.6	100.0
VLA-based Methods						
UniVLA 2025	81.7	96.9	91.1	76.8	91.7	96.7
FSDrive 2025	85.1	98.2	93.8	80.1	93.3	99.9
AutoVLA 2025	89.1	98.4	95.6	81.9	98.0	99.9
DriveVLA-W0 2025b	90.2	98.7	99.1	83.3	95.3	99.3
SpanVLA 2026	90.3	99.1	97.1	86.3	95.2	100.0
ReCogDrive 2025c	90.8	97.9	97.3	87.3	94.9	100.0
SGDrive 2026a	91.1	98.6	97.8	85.8	96.2	100.0
LatentVLA 2026	92.4	98.9	98.2	88.2	96.0	100.0
End-to-End Methods						
DrivingGPT 2024	82.4	98.9	90.7	79.7	94.9	95.6
UniAD 2023	83.4	97.8	91.9	78.8	92.9	100.0
DriveX-S 2025	84.5	97.5	94.0	79.7	93.0	100.0
World4Drive 2025	85.1	97.4	94.3	79.9	92.8	100.0
DRAMA 2025	85.5	98.0	93.1	80.1	94.8	100.0
VADv2 2026	86.2	98.1	94.8	80.6	94.3	100.0
PRIX 2026	87.8	98.1	96.3	82.3	94.1	100.0
DiffusionDrive 2025	88.1	98.2	96.2	82.2	94.7	100.0
DIVER 2026	88.3	98.5	96.5	82.6	94.9	100.0
TrajDiff 2025	88.5	98.1	97.0	82.7	94.3	100.0
Hydra-MDP++ 2025a	91.0	98.6	98.6	85.7	95.1	100.0
iPad 2025	91.7	98.6	98.3	88.0	94.9	100.0
Centaur 2025	92.6	99.5	98.9	85.9	98.0	100.0
DriveSuprim 2025	93.5	98.6	98.6	91.3	95.5	100.0
DriveR 2026	93.7	99.0	98.9	90.0	96.7	100.0
RAP-DINO 2026	93.8	99.1	98.9	90.3	96.7	100.0
RRDrive (Ours)	95.1	99.3	99.0	92.2	97.6	100.0

Table 1: Comparison of results on the NAVSIM benchmark. All metrics are higher-is-better. Best results are highlighted in bold. VLA-based and E2E methods are sorted ascendingly by PDMS. $\uparrow / \downarrow$: higher / lower is better.

identified in the abstract yields a state-of-the-art final score, with the margin driven by the ego-progress (EP) and time-to-collision (TTC) terms while the safety submetrics stay saturated.

Gain on Candidate Generation Quality

Table 3 evaluates the candidate pool—good/bad ratios, mean candidate PDMS, and good candidates per scene—independently of the scorer, isolating the abstract’s first bottleneck (a low candidate-quality floor) and the risk-conditioned generation module built to raise it. On these metrics RiskOcc raises the *good ratio* (fraction of candidates with $q \geq 0.7$) by 8.4 percentage points, nearly halves the *bad ratio* ($q < 0.3$), improves the *mean candidate PDMS* from 0.797 to 0.858, and increases the number of *good candidates per scene* from 51.4 to 56.8.

This reflects a *concentration* rather than a diversification of the candidate budget: among good candidates, the endpoint spread shrinks from 2.81 to 1.55 m and the number of distinct endpoint modes drops from 20.7 to 16.2, i.e. RiskOcc reallocates proposals from infeasible behaviors toward the feasible

Variant	Module	NC $\uparrow$	DAC $\uparrow$	DDC $\uparrow$	TTC $\uparrow$	EP $\uparrow$	Comf. $\uparrow$	PDMS $\uparrow$	Δ
Baseline (no RiskOcc)	–	99.02	98.92	97.23	96.72	89.91	100.00	93.67	–
+ global RiskOcc encoder	Gen.	99.24	99.08	97.51	97.34	90.96	99.99	94.49	+0.82
+ trajectory-conditioned sampling	Rank.	99.30	99.02	97.56	97.57	91.25	99.99	94.68	+0.19
+ bidirectional cross-modal fusion	Gen.	99.37	99.04	97.46	97.78	91.51	100.00	94.90	+0.22
+ TTC/EP & RiskOcc refine	Rank.	99.34	98.98	97.41	97.55	92.19	100.00	95.06	+0.16

Table 2: Cumulative ablation on navtest (argmax, no re-ranking), decomposed into all PDMS sub-metrics. **Module** tags the *primary* pathway each component serves: Gen. = candidate generation, Rank. = candidate ranking. All metrics are higher-is-better; NC = no at-fault collision, DAC = drivable-area compliance, DDC = driving-direction compliance, TTC = time-to-collision, EP = ego-progress, Comf. = comfort. Δ is the absolute PDMS gain over the preceding row. Rows are model-lineage proxies: each step carries minor co-changes, so tags denote the dominant pathway rather than an exclusive one.

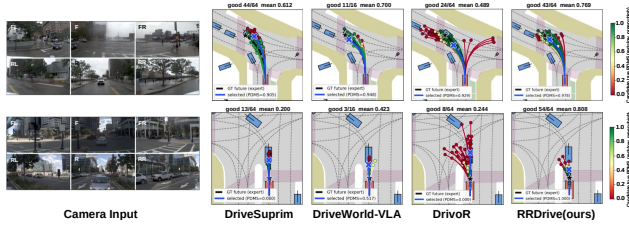


Figure 5: Qualitative comparison of candidate generation on two generation-challenging scenes. Compared with DrivoR (third column), RRDrive (fourth column) concentrates the finite proposal budget within feasible, high-quality regions aligned with the expert trajectory, producing more good candidates and higher mean candidate PDMS.

high-quality region. The qualitative comparison in Figure 5 confirms that RRDrive overcomes the low-quality candidate-generation failure by concentrating the finite proposal budget within feasible, high-quality motion regions rather than dispersing it across unsafe or ineffective branches. In the two generation-challenging scenes, the fourth column produces compact candidate bundles aligned with the expert trajectory, yielding 43/64 and 54/64 good candidates with mean PDMS values of 0.769 and 0.808, whereas the DrivoR candidates in the third column diverge toward multiple low-score branches and contain only 26/64 and 8/64 good candidates with mean PDMS values of 0.489 and 0.244, respectively.

Metric	Baseline	Ours	$\Delta(\%)$
Good ratio $\uparrow$	80.3%	88.7%	+10.5
Bad ratio $\downarrow$	13.6%	7.6%	−44.1
Mean cand. PDMS $\uparrow$	0.797	0.858	+7.7
Good cand./scene $\uparrow$	51.4	56.8	+10.5

Table 3: Candidate generation quality. $\uparrow/\downarrow$: higher / lower is better; $\Delta(\%)$ is the relative change over the baseline.

Gain on Candidate Ranking

Table 4 assesses how the scorer orders the 64 candidates within each scene—scene-wise Spearman between predicted and true PDMS, oracle recall@ K (whether the true optimum lies in the top K), oracle rank of the true optimum (lower is

better), oracle gap (PDMS shortfall of the selected trajectory relative to the in-scene optimum), and the selected PDMS—directly targeting the abstract’s second bottleneck, unreliable fine-grained ranking, which the trajectory-conditioned local risk pathway is built to fix. RiskOcc improves every metric: the scene-wise Spearman correlation rises from 0.407 to 0.584, oracle recall@1 from 5.1% to 15.4% (3.0 $\times$), the oracle rank from 33.9 to 27.7, and the selected PDMS from 0.937 to 0.951. To probe the underlying mechanism, Table 5 predicts the good-candidate order from geometry, risk, or both under scene-grouped cross-validation: the risk profile contributes substantial ranking information beyond geometry (0.37 $\rightarrow$ 0.53), whereas risk alone is weak. This is consistent with the weak linear correlation between simple risk statistics and PDMS ($|r| < 0.08$) and the comparable global scorer-PDMS Pearson of the two models (0.57 vs. 0.56): RiskOcc functions as nonlinear evidence for the scorer rather than a linear surrogate.

Metric	Baseline	Ours	$\Delta(\%)$
Scene Spearman $\uparrow$	0.407	0.584	+43.5
Oracle recall@1 $\uparrow$	5.1%	15.4%	+202.0
Oracle recall@5 $\uparrow$	12.2%	29.3%	+140.2
Oracle rank $\downarrow$	33.9	27.7	−18.3
Oracle gap $\downarrow$	0.057	0.034	−40.4
Selected PDMS $\uparrow$	0.937	0.951	+1.5

Table 4: Scene-level ranking quality. $\uparrow/\downarrow$: higher / lower is better; $\Delta(\%)$ is the relative improvement over the baseline.

Feature set	Ego-progress $\uparrow$	True PDMS $\uparrow$
Geometry only	0.369	0.371
RiskOcc only	0.174	0.177
Geometry + RiskOcc	0.535	0.533

Table 5: Within-good-candidate ranking under scene-grouped cross-validation (mean scene-wise Spearman $\uparrow$): the risk profile adds ranking information beyond trajectory geometry.

The qualitative comparison in Figure 6 confirms that RRDrive resolves the fine-grained ranking failure by identifying the best trajectory within dense bundles of geometri-

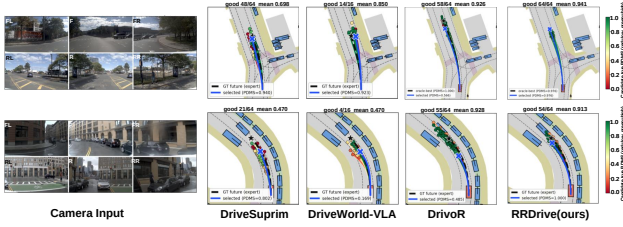


Figure 6: Qualitative comparison of candidate ranking. Within dense bundles of geometrically similar, high-quality candidates, RRDrive selects a near-optimal trajectory where the baseline (third column) mis-ranks and selects a low-scoring one, isolating the improvement to ranking rather than to candidate generation.

cally similar, high-quality candidates. In the upper scene, the baseline method in the third column already generates 58/64 good candidates with a mean PDMS of 0.926 and contains an oracle trajectory with PDMS 1.000, yet selects a trajectory scoring only 0.566, while RRDrive selects a 0.976 trajectory; more decisively, in the lower scene, DrivoR has slightly better pool statistics than RRDrive (55/64 good candidates and mean PDMS 0.928 versus 53/64 and 0.913) but selects a 0.485 trajectory, whereas RRDrive selects the 1.000 trajectory, isolating the improvement to candidate ranking rather than candidate generation.

Difficulty-Stratified Analysis

Table 6 partitions the scenes into terciles by baseline difficulty and reports, for the hardest tercile of each axis, candidate PDMS and good ratio (generation), scene Spearman and selected PDMS (ranking), and in-good Spearman and oracle recall@1 (confusion), testing the abstract’s claim that the gains land where the baseline is weakest. They do, concentrating in the hardest cases along both axes. In the hardest *generation* tercile (lowest baseline candidate PDMS), pool quality improves most; in the hardest *ranking* tercile (largest baseline oracle gap) and the hardest *confusion* tercile (good candidates most similar in quality), the ranking metrics improve most, with the within-good Spearman rising from 0.265 to 0.674 and oracle recall@1 from 3.6% to 24.9%.

Scenario	Metric $\uparrow$	Baseline	Ours	$\Delta(\%)$
Gen-hard	Candidate PDMS	0.540	0.703	+30.2
Gen-hard	Good ratio	50.1%	69.8%	+39.3
Rank-hard	Scene Spearman	0.072	0.497	+590.3
Rank-hard	Selected PDMS	0.816	0.884	+8.3
Confu-hard	In-good Spearman	0.265	0.674	+154.3
Confu-hard	Oracle recall@1	3.6%	24.9%	+591.7

Table 6: Hardest-tercile analysis (baseline vs. Ours). All listed metrics are higher-is-better ($\uparrow$); $\Delta(\%)$ is the relative improvement over the baseline.

Ablation Study

Table 2 decomposes each RiskOcc component under a strict argmax protocol (no re-ranking), reporting the full PDMS sub-metric breakdown. Starting from the reproduced DrivoR baseline (PDMS 0.9367), we add components cumulatively and tag each by its primary pathway—generation (Gen.) or ranking (Rank.).

Generation vs. ranking contributions. The two generation-oriented components—global risk encoding and bidirectional cross-modal fusion—raise the score by +0.0104, while the two ranking-oriented components—trajectory-conditioned risk sampling and the TTC/EP & risk-representation refinement—add a further +0.0035, reaching **0.9506** for a total gain of +0.0139. Consistent with our two design goals, conditioning generation on the global RiskOcc field yields the larger share by improving the candidate pool, whereas the trajectory-aligned risk profile contributes a smaller but complementary gain by sharpening selection among already-strong candidates.

Where the gain comes from. The improvement concentrates in ego-progress (EP 0.8991 $\rightarrow$ 0.9219, +0.0228) and time-to-collision (TTC 0.9672 $\rightarrow$ 0.9755), while the safety-critical sub-metrics remain saturated throughout (NC $\geq$ 0.992, DAC $\geq$ 0.989, DDC $\geq$ 0.972, Comfort = 1.000). The gain therefore stems from more accurate progress- and TTC-aware generation and selection rather than from conservative, progress-sacrificing behavior—precisely the fine progress–risk trade-off that an explicit, trajectory-aligned risk signal is designed to resolve.

Robustness to Lossy RiskOcc

To test whether the framework depends on oracle risk, we train an external RiskOcc prediction module that infers the risk modality from sensors, yielding a lossy RiskOcc field in place of the ground truth. As the scorer is trained on clean risk, we bridge the train/test discrepancy via a lightweight mixed-source adaptation over ground-truth and predicted RiskOcc. With fully predicted risk, RRDrive attains a PDMS of 0.9400, clearly exceeding the no-RiskOcc baseline (0.9367) while trailing the oracle-risk upper bound (0.9506). The residual gap stems mainly from our perception module, whose multi-step temporal forecasting of RiskOcc remains limited; improving this prediction quality—where substantial headroom remains—would directly benefit the fusion framework. These gains under lossy, sensor-derived risk confirm that they arise from the risk-aware formulation rather than from privileged access to oracle risk.

Conclusion

We identify compact scene representation as a shared bottleneck of multimodal trajectory generation and selection, and address it with risk-aware occupancy (RiskOcc), a dense, temporally aligned, trajectory-queryable field that unifies static, lane, and dynamic risks. Built on this representation, RRDrive conditions candidate generation on the global risk field and augments the scorer with a trajectory-aligned local risk profile, while RiskOcc-NAVSIM supplies the required supervision through automatic annotation. RRDrive raises

candidate quality and ranking reliability, especially in the hardest scenes, and attains a state-of-the-art PDMS of 0.951. An external predictor further verifies that such performance gains persist under sensor-derived risk. The main remaining limitation lies in the multi-step temporal forecasting of dynamic RiskOcc, and improving its prediction quality is a promising direction to fully unlock the representation’s benefits for end-to-end driving.

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